
Numerical integration in the restricted three-body problem

Building a tool for studying motion in the Earth–Moon system



“Start in Science” Olympiad

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Introduction

The three-body problem

A classical problem
of celestial mechanics.
No analytical solution
exists

CR3BP

Two massive bodies
move on circular orbits.
The third body has
negligible mass

Rotating frame

The Earth and Moon are
stationary.
Coriolis and centrifugal
accelerations appear

The circular restricted three-body problem is the basic model for studying dynamics in the Earth–Moon system

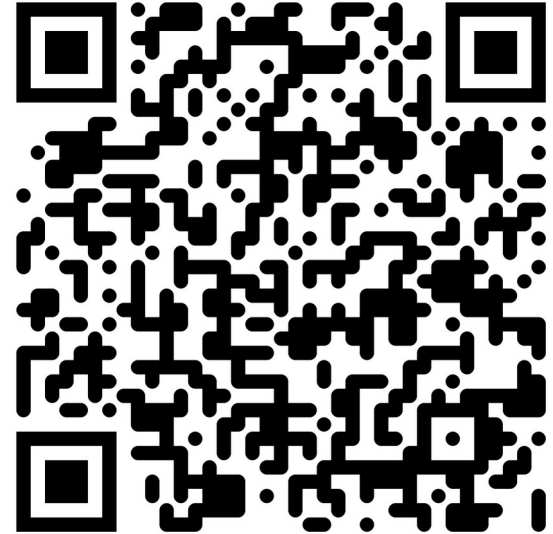
Relevance and existing tools

The CR3BP is the basic model of Earth–Moon dynamics. No analytical solution exists

Tool	CR3BP	3D	Thrust	Jacobi	Web
GMAT (NASA)	●	●	●	—	—
STK (Ansys/AGI)	—	●	●	—	—
Universe Sandbox	—	●	●	—	—
PhET Simulations	—	—	—	—	●
Desmos	—	—	—	—	●
“Orbit. Challenge”	—	●	—	—	●
rocketlauncher.space	●	●	●	●	●

I looked for an existing simulator and found none

<https://rocketlauncher.space>



- 3D visualisation of the Earth–Moon system
- Engine with a configurable thrust window
- Three integrators with Jacobi-based accuracy control
- Lagrange points L1–L5

Aim and objectives

Building an educational web simulator for trajectories in the Earth-Moon system

- 1 Study the theoretical foundations of the CR3BP
- 2 Derive the equations of motion in a rotating frame
- 3 Implement trajectory computation using the Euler method
- 4 Implement the Verlet method and compare its accuracy with Euler
- 5 Find the Lagrange points L1-L5 numerically
- 6 Create an interactive 3D web simulator

Mathematical model

Equations of motion in the rotating frame:

$$\ddot{x} = 2\omega\dot{y} + \omega^2 x - \frac{GM_E(x + d_E)}{r_E^3} - \frac{GM_M(x - d_M)}{r_M^3} + \frac{F_x}{m}$$

$$\ddot{y} = -2\omega\dot{x} + \omega^2 y - \frac{GM_E \cdot y}{r_E^3} - \frac{GM_M \cdot y}{r_M^3} + \frac{F_y}{m}$$

$$\ddot{z} = -\frac{GM_E \cdot z}{r_E^3} - \frac{GM_M \cdot z}{r_M^3} + \frac{F_z}{m}$$

G

grav. constant

M_e, M_m

body masses

d_e, d_m

from the
barycentre

r_e, r_m

to the centres of
mass

ω

angular velocity

m = M - λt

fuel mass

2ωẏ and -2ωẋ — the Coriolis acceleration. It depends on velocity

+ thrust model: $F = \text{const}$ within the window $[t_{\text{on}}, t_{\text{off}}]$

Scope of applicability

01

Circular orbit

The Moon's orbit is treated as circular
(actual eccentricity = 0.055)

02

Negligible spacecraft mass

The spacecraft's mass is negligible
compared with the Earth and the Moon

03

No Sun

The attraction of the Sun and other
bodies is not taken into account

The model is valid for qualitative study
of the dynamics and for educational use

Real missions are designed with
full ephemeris models

Three integrators

The Euler method

Simple, order $O(h)$

Does not conserve energy

$O(h)$

Verlet with explicit Coriolis handling

Symplectic, $O(h^2)$ in theory

But Coriolis spoils it

Degrades to $O(h)$ in the CR3BP

*$O(h)$ in the
CR3BP*

Verlet with iterative velocity correction

3 iterations per step refine v

Restores order $O(h^2)$

$O(h^2)$

The Coriolis acceleration $2\omega \times v$ depends on velocity $\rightarrow$ Verlet with explicit Coriolis handling loses its order $\rightarrow$ iteration restores $O(h^2)$

The Jacobi integral

The only conserved quantity in the CR3BP

$$C_J = 2U - V^2$$

$e = \frac{C_J(t) - C_J(0)}{C_J(0)}$ - measure of accumulated error

< 0.001%

Excellent

< 0.01%

Good

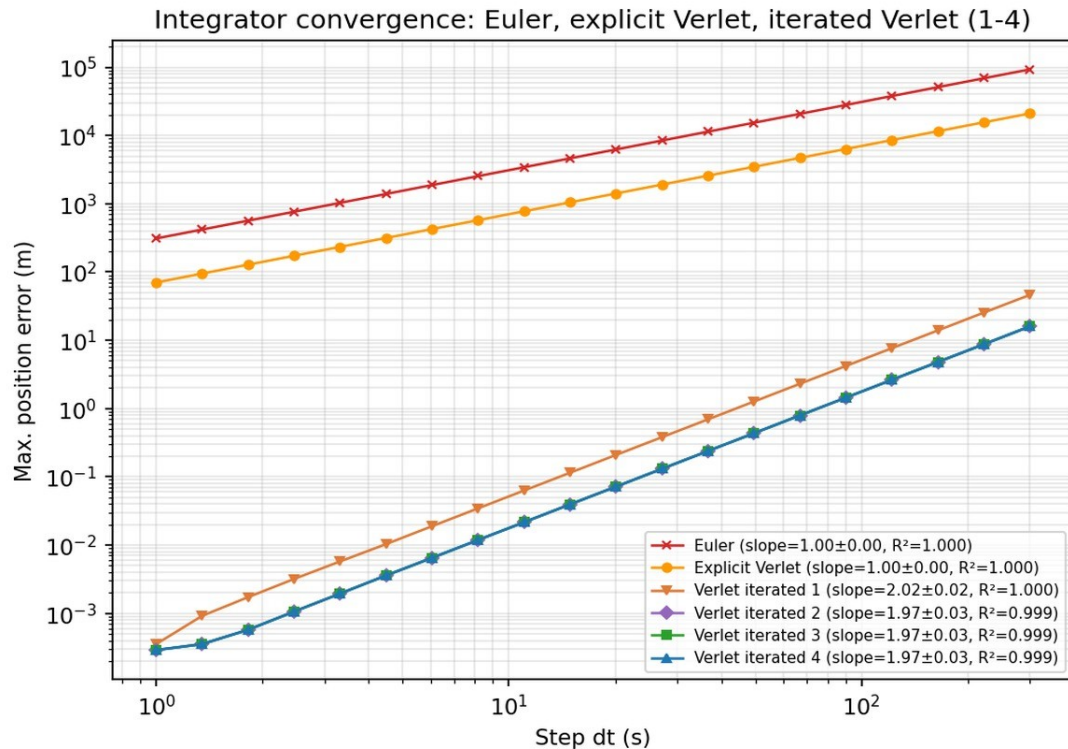
< 0.1%

Acceptable

> 0.1%

Poor

Log-log convergence



Euler
slope 1.00 $\rightarrow \mathcal{O}(h)$

Explicit Verlet
slope 1.00 $\rightarrow \mathcal{O}(h)$

Iterative Verlet
slope 1.97 $\rightarrow \mathcal{O}(h^2)$

$dt = 30 \text{ s} \mid R^2 = 0.999$

Euler: 8,400 m

Explicit Verlet: 1,900 m

Iterative Verlet: 0.13 m

64,600× more accurate in position

Adaptive step

Near a body $1/r$ grows $\rightarrow$ a smaller step is needed

$$dt = \frac{dt_{\max}}{1 + \frac{v}{v_{\text{ref}}} \left(\frac{dt_{\max}}{dt_{\min}} - 1 \right)}$$

Far from the bodies

$$dt = dt_{\max}$$

Forces are small, the step is at its maximum



Transition zone

dt decreases
in proportion to v



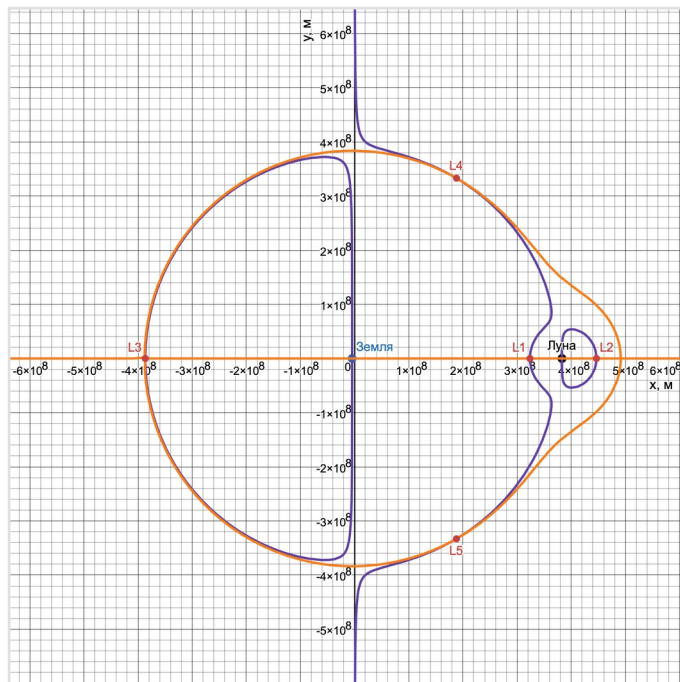
Near a body

$$dt \approx dt_{\min}$$

Forces are large, the step is at its minimum

Locating the Lagrange points: Desmos

Graphical analysis of the total acceleration in Desmos



$$\omega^2 x - \frac{GM_E(x+d_E)}{((x+d_E)^2+y^2)^{3/2}} - \frac{GM_M(x-d_M)}{((x-d_M)^2+y^2)^{3/2}} = 0$$

$$y \cdot \left(\frac{GM_E}{((x+d_E)^2+y^2)^{3/2}} + \frac{GM_M}{((x-d_M)^2+y^2)^{3/2}} - \omega^2 \right) = 0$$

$$z = 0$$

From graph to algorithm

The graphical investigation led to formalising the bisection method for locating the Lagrange points automatically

Lagrange points L1–L5

5 equilibrium points of the Earth–Moon system

Point	x, km	y, km	Method	Resid. accel., m/s ²
L1	323 696	0	Bisection	$< 10^{-18}$
L2	446 531	0	Bisection	$< 10^{-18}$
L3	−386 651	0	Bisection	$< 10^{-18}$
L4	188 081	332 925	Newton	$< 10^{-18}$
L5	188 081	−332 925	Newton	$< 10^{-18}$

Desmos graph → sign change → bisection → accuracy $< 10^{-18}$ m/s²

Analytical approximations (Hill sphere) give a relative error of 0.4–0.8% — the numerical solution is more accurate

Personal contribution

Physics

Derivation of the CR3BP equations of motion in a rotating frame

Validation

Implementation of the Jacobi integral as an accuracy-checking tool

Num. methods

3 integrators; discovering the degradation of Verlet and implementing the variant with iterative velocity correction

Experiments

A series of 7 computational experiments for quantitative verification

Algorithms

Locating the Lagrange points L1–L5: bisection + Newton's method

Development

3D web simulator + scripts for finding initial conditions

Directions for development

Elliptical orbit

Accounting for the Moon's eccentricity

Solar gravity

Extension to the four-body problem

Radiation pressure

Accounting for solar radiation pressure

Station keeping

Holding the spacecraft near a halo orbit

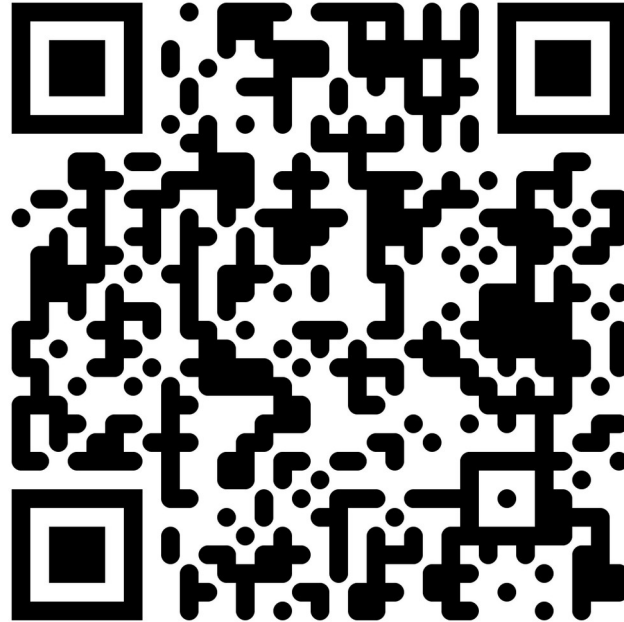
Laboratory exercises

A set of assignments built on the simulator

Conclusions

- ✓ The theoretical foundations of the CR3BP were studied and the equations of motion derived
- ✓ 3 integrators were implemented. Verlet with iterative velocity correction is 64,600× more accurate than Euler
- ✓ The Lagrange points L1–L5 were found numerically (resid. accel. $< 10^{-18}$)
- ✓ The model was verified by a series of 7 numerical experiments
- ✓ An educational 3D web simulator was built and published online

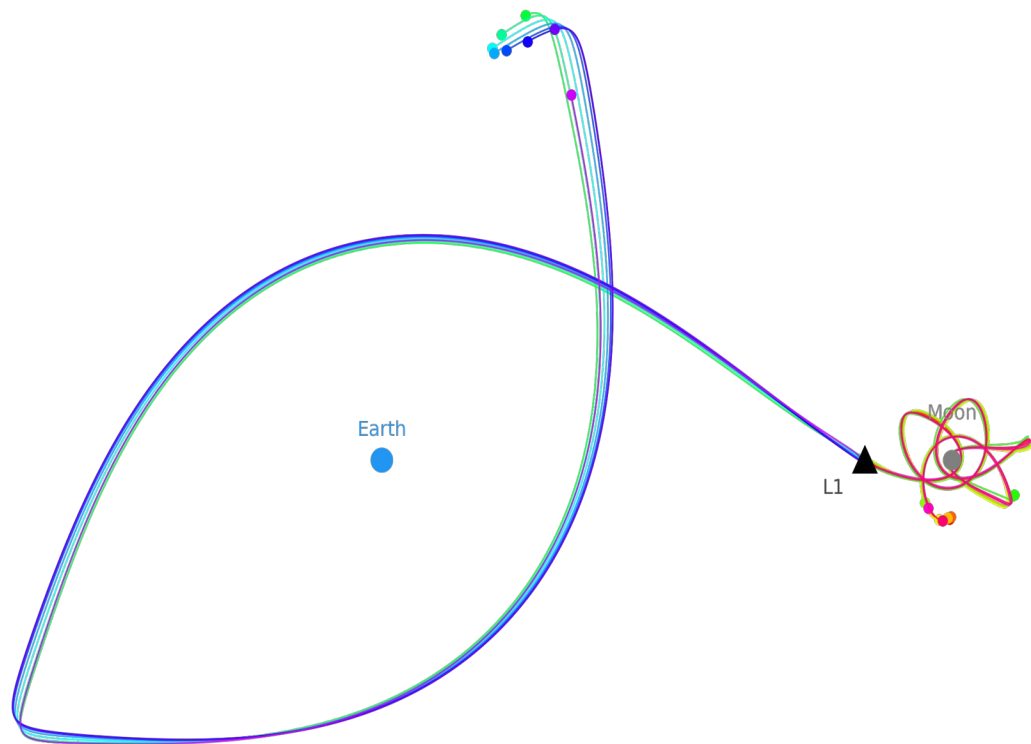
Thank you for your attention



Appendix

Supplementary material for answering questions

Chaos at L1



Δv_0

10 m/s

Lyapunov exponent λ

0.64 day^{-1}

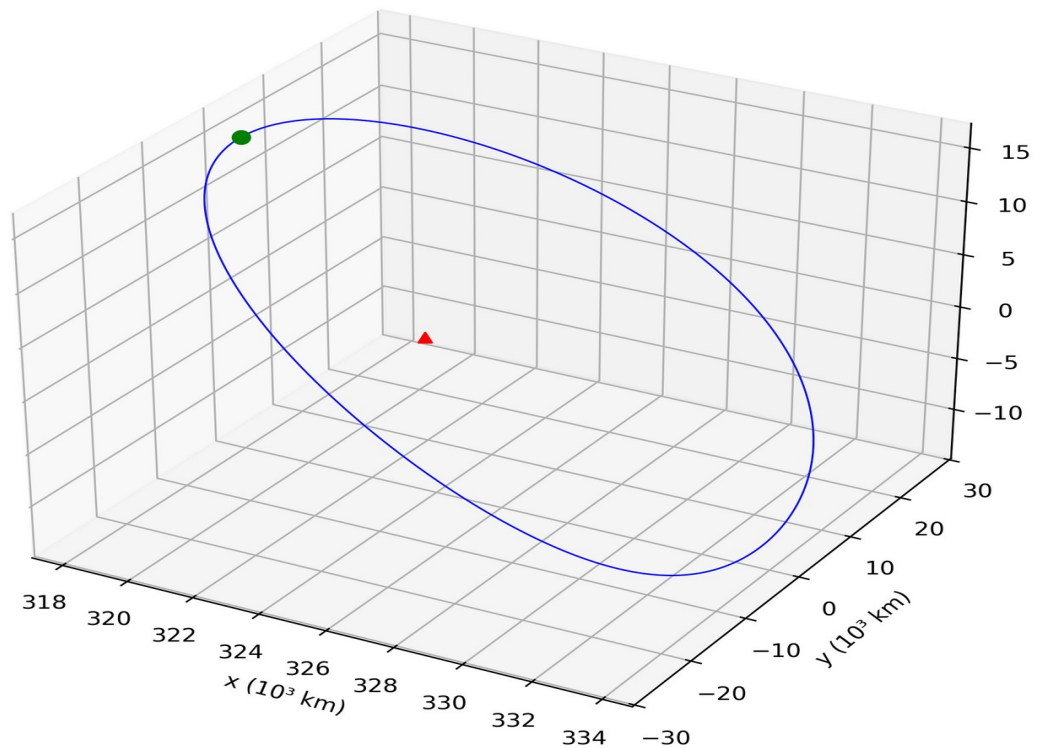
Predictability horizon

~2 days

Directions / time

16 / 30 days

Halo orbit at L1



Amplitude Az

15,000 km

Period

293 h (12.2 days)

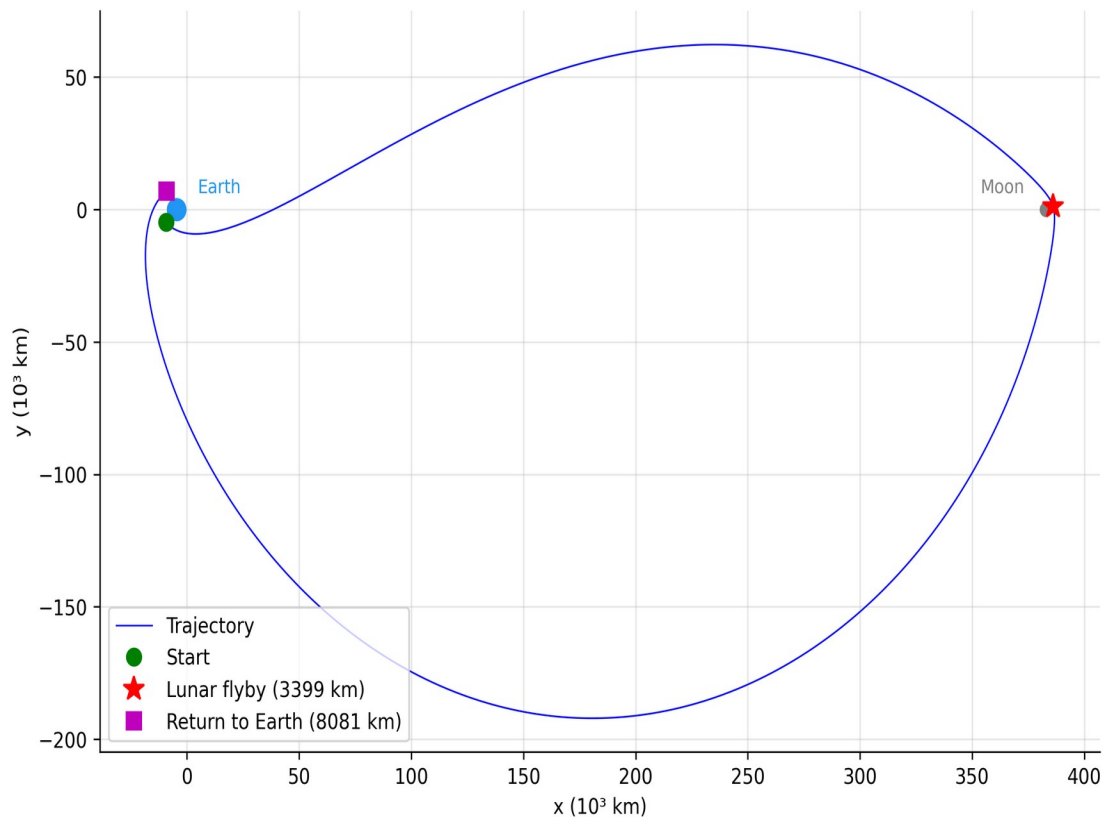
Closure error

3.6 km

Jacobi drift

3.8×10^{-11}

Free return



Lunar flyby

3,399 km

Return to Earth

8,081 km

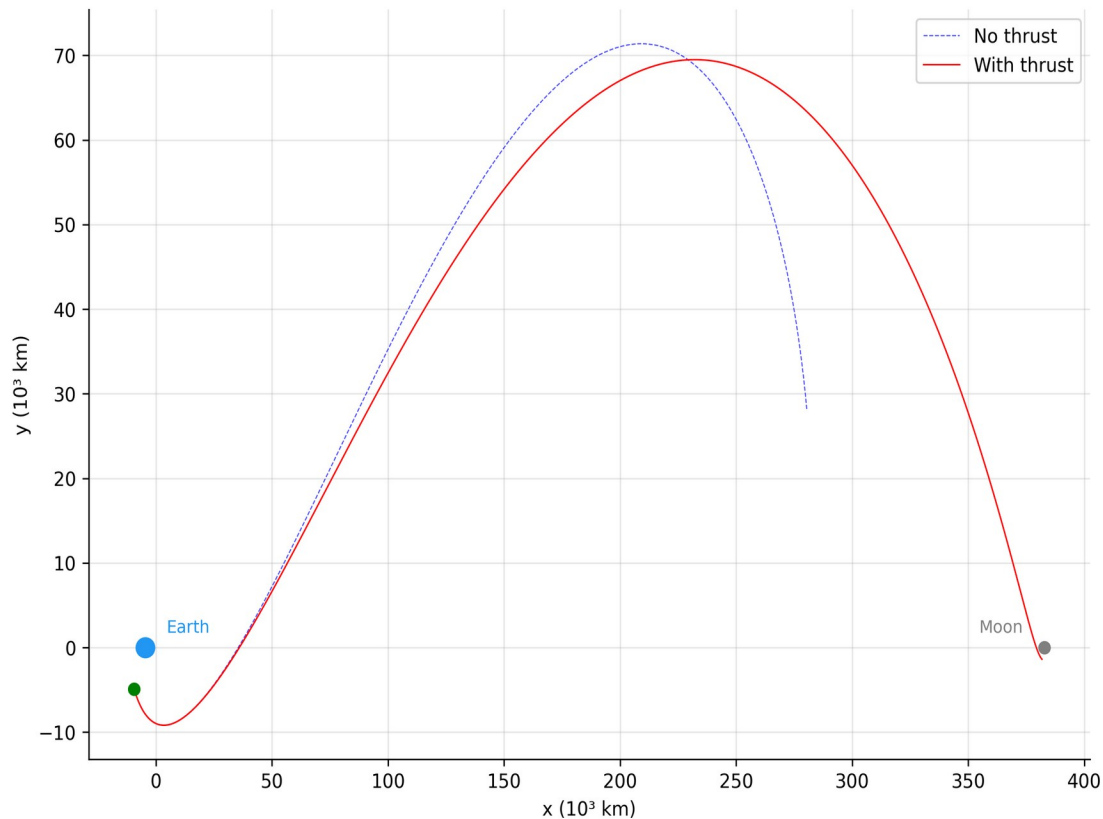
Flight time

168 h

+1 m/s $\rightarrow \Delta r$ Moon

+89 km

Powered transfer



Thrust / duration

5 N / 3 h

Δv

108 m/s

Lunar flyby

2,494 km

Fuel

18 kg (3.6%)

Integrator comparison

Jacobi integral drift, halo orbit at L1, $dt = 30$ s, $T = 720$ h

Integrator	Mode	Max. drift
Euler	fixed	2.2×10^{-6}
Verlet half-step	fixed	1.1×10^{-6}
Verlet iterated	fixed	3.8×10^{-11}

58 000×

by Jacobi drift

64 600×

by position error

A ready-made tool for laboratory work

The simulator is ready for classroom use — no installation, runs in the browser

01

Halo orbits

Parametric study:
period, closure error,
Jacobi drift

02

Free return

Sensitivity to initial
conditions, lunar flyby

03

Chaos and Lyapunov

Predictability horizon,
divergence of
trajectories

04

Thrust optimisation

Choosing the engine
firing window for a
manoeuvre

Browser access · No installation · 7 ready-made experiments · Open source

Technical stack

3D simulator

JavaScript · Three.js
HTML · CSS
~1,500 lines

Experiments

Python · NumPy
SciPy (RK45 for
validation)
Matplotlib · 7 scripts

Deployment

VPS Beget.com
1 vCPU · 1 GB RAM
NVMe SSD · 210
₽/month

Tools

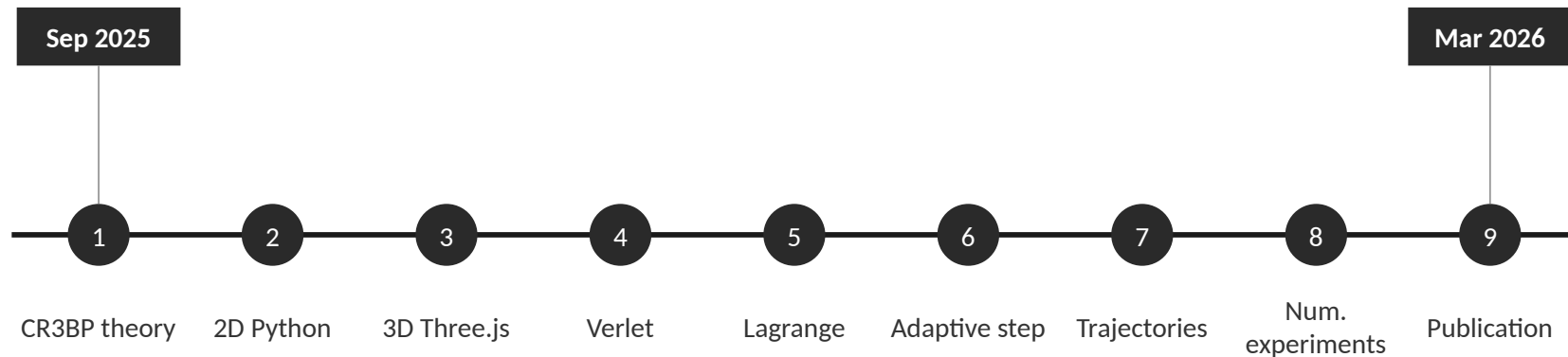
Git · GitHub
Desmos (exploring L1–
L5)
Python ~2,500 lines

All code written from scratch · Three.js + Python + NumPy · VPS at 210 ₽/month

References

- [1] V. V. Beletsky — Essays on the Motion of Celestial Bodies
- [2] G. N. Duboshin — Celestial Mechanics: Principal Problems and Methods
- [3] D. E. Okhotsimsky, Yu. G. Sikharulidze — Fundamentals of Space Flight Mechanics
- [4] A. A. Tselousova — Numerical-Analytical Methods for Constructing Trajectories in the Three- and Four-Body Problems
- [5] Dmitri Rozmanov, Peter G Kusalik - «Robust rotational-velocity-Verlet integration methods»

Project timeline



7 months: from theory to publication on the web

Derivation of the equations of motion

The equations of motion expressed via the effective potential:

$$\Omega(x, y, z) = \frac{1}{2} \omega^2(x^2 + y^2) + \frac{GM_E}{r_E} + \frac{GM_M}{r_M}$$

$$\ddot{x} = 2\omega\dot{y} + \frac{\partial\Omega}{\partial x}$$

$$\ddot{y} = -2\omega\dot{x} + \frac{\partial\Omega}{\partial y}$$

$$\ddot{z} = \frac{\partial\Omega}{\partial z}$$

Ω

eff. potential

ω

angular velocity

$\mathbf{r}_e, \mathbf{r}_m$

to the centres
of mass

$\mathbf{d}_e, \mathbf{d}_m$

from the
barycentre

G

grav. constant

M_e, M_m

body masses

$2\omega\dot{y}$ and $-2\omega\dot{x}$ — the Coriolis terms; they are not part
of Ω